KAIZEN MATHS

Lesson starter

Powers of 10 and Standard Form

Standard Form Operations

Name

S1

Express 6995904 in standard form

S2

Evaluate 7.95×10^5

S3

$(4.2 \times 10^5) \times (5.096 \times 10^1)$

S4

Express 0.05946 in standard form

Show your working

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NUM-F1AA-L3 · S

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Exit ticket

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Name

E1

Evaluate 4.2076×10^{-4}

E2

$(6.7 \times 10^{5}) \times (2.06 \times 10^{3})$

E3

Express 343444 in standard form

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Exit ticket

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Teacher answers & working

Powers of 10 and Standard Form · Standard Form Operations

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S1

Express 6995904 in standard form

Answer:

6.995904×10^{6}

Working:

Step 1: Standard form requires the coefficient to be at least 1 and less than 10

Step 2: Count the place-value shifts needed to form the coefficient

6995904 → digits shift 6 places right → 6.995904

Step 3: Determine the power of 10

The coefficient was made by shifting digits 6 places right, so the compensating power is 6.

Step 4: Write in standard form

$6995904 = 6.995904 \times 10^{6}$

S2

Evaluate 7.95×10^{5}

Answer:

795000

Working:

Step 1: Identify the coefficient and exponent

Coefficient: 7.95, Exponent: 5

Step 2: Shift the digits using the power of 10

10^{5} shifts the digits 5 places left while the decimal point stays fixed.

7.95 → digits shift 5 places left → 795000

Step 3: Write the final answer

$7.95 \times 10^{5} = 795000$

Teacher answers & working

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S3

$(4.2 \times 10^5) \times (5.096 \times 10^1)$

Answer:

2.14032×10^7

Working:

Step 1: Separate coefficients and powers of 10

$(4.2 \times 5.096) \times (10^5 \times 10^1)$

Step 2: Calculate the coefficients

$4.2 \times 5.096 = 21.4032$

Step 3: Apply index laws for powers of 10

$10^5 \times 10^1 = 10^6$

Step 4: Combine the coefficient and power of 10

21.4032×10^6

Step 5: Ensure coefficient is at least 1 and less than 10

Final answer: 2.14032×10^7

S4

Express 0.05946 in standard form

Answer:

5.946×10^{-2}

Working:

Step 1: Standard form requires the coefficient to be at least 1 and less than 10

Step 2: Count the place-value shifts needed to form the coefficient

0.05946 → digits shift 2 places left → 5.946

Step 3: Determine the power of 10

The coefficient was made by shifting digits 2 places left, so the compensating power is -2.

Step 4: Write in standard form

$0.05946 = 5.946 \times 10^{-2}$

Teacher answers & working

Powers of 10 and Standard Form · Standard Form Operations

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E1

Evaluate 4.2076×10^{-4}

Answer:

0.000421

Working:

Step 1: Identify the coefficient and exponent

Coefficient: 4.2076, Exponent: -4

Step 2: Shift the digits using the power of 10

10^{-4} shifts the digits 4 places right while the decimal point stays fixed.

4.2076 → digits shift 4 places right → 0.000421

Step 3: Write the final answer

$4.2076 \times 10^{-4} = 0.000421$

E2

$(6.7 \times 10^{5}) \times (2.06 \times 10^{3})$

Answer:

1.3802×10^{9}

Working:

Step 1: Separate coefficients and powers of 10

$(6.7 \times 2.06) \times (10^{5} \times 10^{3})$

Step 2: Calculate the coefficients

$6.7 \times 2.06 = 13.802$

Step 3: Apply index laws for powers of 10

$10^{5} \times 10^{3} = 10^{8}$

Step 4: Combine the coefficient and power of 10

13.802×10^{8}

Step 5: Ensure coefficient is at least 1 and less than 10

Final answer: 1.3802×10^{9}

Teacher answers & working

Powers of 10 and Standard Form · Standard Form Operations

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E3

Express 343444 in standard form

Answer:

3.43444×10^5

Working:

Step 1: Standard form requires the coefficient to be at least 1 and less than 10

Step 2: Count the place-value shifts needed to form the coefficient

343444 → digits shift 5 places right → 3.43444

Step 3: Determine the power of 10

The coefficient was made by shifting digits 5 places right, so the compensating power is 5.

Step 4: Write in standard form

$343444 = 3.43444 \times 10^5$